When Nursing Hours Become the Constraint

A Thought Experiment for Health System Leaders

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Request for Feedback

This paper presents a hypothesis, not a conclusion. The author seeks corrections, challenges, and suggestions from those interested in improving healthcare.

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1 Introduction

Health systems face a familiar set of pressures: thin operating margins, rising costs, and persistent workforce challenges. Among these workforce challenges, the nursing shortage stands out. Vacancy rates remain elevated, turnover is costly, and competition for qualified nurses has driven labor expenses to unprecedented levels.[1]

This paper explores a question that may seem obvious once stated but is rarely examined systematically: *What if available nursing hours—not beds, not operating room time, not equipment—were the binding constraint on health system throughput?*

If nursing capacity is indeed the constraint, it would have profound implications for how health systems evaluate service lines, allocate resources, and make strategic investments. Conventional financial analysis might be pointing leadership in exactly the wrong direction.

This is presented as a hypothesis worth examining, not as settled truth. The author has not had the opportunity to test this framework against real health system data. The goal of this paper is to describe what a nursing-constrained system would look like, explain why conventional accounting would not reveal it, and invite thoughtful discussion on whether the hypothesis has merit.

2 A Different Lens: Constraint-Based Thinking

Every system has at least one constraint—a resource that limits overall output. In a factory, it might be a particular machine. In a restaurant, it might be kitchen capacity during peak hours. In a health system, it could be any number of things: inpatient beds, operating room hours, specialist physician availability, imaging equipment, or nursing staff.

The constraint determines system throughput. No matter how much excess capacity exists elsewhere, the system cannot produce more than the constraint allows. This observation, central to the Theory of Constraints developed by Dr. Eliyahu Goldratt,[2] leads to a counterintuitive insight about profitability.

2.1 The Key Insight

When a resource is constrained, the relevant measure of profitability is not margin per case or margin per service line. It is **throughput per unit of the constrained resource consumed**.

A service line with high margin per case but heavy consumption of the constrained resource may actually be less valuable than a service line with modest margin but light constraint consumption. The high-margin service "uses up" the constraint, leaving less capacity for other work. The modest-margin service leaves constraint capacity available, enabling higher total system throughput.

This is not intuitive. It contradicts how most of us learned to think about profitability. But when a constraint is binding, the math is unambiguous.

[1] According to NSI Nursing Solutions, the 2024 RN vacancy rate stood at 9.6%, representing an average of 47 nursing vacancies per hospital. Hospital labor expenses increased 37% per patient from 2019 to 2022, driven largely by a rise in contract labor from 2% to 11% of total labor expenses. See Kaufman Hall, "National Hospital Flash Report," May 2022.

[2] Goldratt, Eliyahu M. *The Goal: A Process of Ongoing Improvement*. North River Press, 1984. The Theory of Constraints has since been applied across manufacturing, project management, and services industries. For an overview, see https://www.tocinstitute.org/theory-of-constraints.html

2.2 How This Differs from Conventional Analysis

Conventional service line profitability analysis asks: *What is the contribution margin of each service line?* Services are ranked by margin, and strategic decisions follow: invest in high-margin services, scrutinize or divest low-margin ones.

Constraint-based analysis asks a different question: *What is the contribution margin per unit of the constraint consumed?* This can produce dramatically different rankings—and therefore dramatically different strategic conclusions.

The next section illustrates this with a concrete example.

3 A Numerical Example

Consider a hypothetical community health system with eight service lines. Table 1 shows each service line's revenue per case, variable costs, contribution margin, and average nursing hours required per case.

Table 1: Service Line Financial and Operational Data (Hypothetical)

Service Line	**Revenue per Case**	**Variable Costs**	**Contribution Margin**	**Nursing Hours**	**Margin per Nursing Hr**
Complex Cardiac	$85,000	$52,000	$33,000	144	$229
Neurosurgery	$62,000	$38,000	$24,000	96	$250
Total Joint (TKA/THA)	$28,000	$16,000	$12,000	28	$429
General Surgery	$18,000	$10,500	$7,500	24	$313
Labor & Delivery	$12,000	$7,200	$4,800	18	$267
Same-Day Surgery	$8,500	$4,800	$3,700	5	$740
Oncology Infusion	$4,200	$2,800	$1,400	3	$467
Observation/Short Stay	$3,200	$1,900	$1,300	6	$217

3.1 Two Different Rankings

Ranked by conventional contribution margin per case:

1. Complex Cardiac — $33,000
2. Neurosurgery — $24,000
3. Total Joint — $12,000
4. General Surgery — $7,500
5. Labor & Delivery — $4,800
6. Same-Day Surgery — $3,700
7. Oncology Infusion — $1,400
8. Observation — $1,300

Ranked by contribution margin per nursing hour:

1. Same-Day Surgery — $740/hr
2. Oncology Infusion — $467/hr
3. Total Joint — $429/hr
4. General Surgery — $313/hr
5. Labor & Delivery — $267/hr

6. Neurosurgery — $250/hr
7. Complex Cardiac — $229/hr
8. Observation — $217/hr

Complex Cardiac drops from #1 to #7. Same-Day Surgery rises from #6 to #1.

3.2 What Happens When You Expand the "Wrong" Service

Suppose leadership, looking at conventional margins, decides to invest in expanding the Complex Cardiac program. Marketing is increased, another surgeon is recruited, and volume grows by 50 cases per year.

On paper, this looks excellent: 50 cases × $33,000 margin = $1.65 million in additional contribution.

But each cardiac case requires 144 nursing hours. Those 50 additional cases consume 7,200 nursing hours—hours that must come from somewhere when nursing is constrained.

If those nursing hours were previously serving Same-Day Surgery cases:

- 7,200 hours ÷ 5 hours per case = 1,440 same-day cases displaced
- Lost contribution: 1,440 × $3,700 = $5.33 million

Net effect of the "profitable" expansion: negative $3.68 million.

This loss does not appear cleanly in financial reports. What leadership sees: Cardiac revenue up, cardiac margin up, same-day volume "soft" (attributed to competition or seasonality), overall hospital margins mysteriously declining, nursing overtime increasing as staff attempt to do both.

The structural cause—constraint consumption—remains invisible.

4 What You Would Expect to See: Operational Signs

If nursing hours were the binding constraint in a health system, certain operational patterns would be observable. These are not proof of a nursing constraint, but their presence would be consistent with the hypothesis.

Beds closed for staffing, not physical capacity. The system reports available bed capacity, but units are frequently on divert or unable to accept admissions because nurses are not available to staff the beds. Physical plant is not the limitation.

Emergency department holds and boarding. Patients wait in the ED for inpatient beds despite census below physical capacity. The bottleneck is not bed availability but the nursing staff required to accept and care for new admissions on the floors.

Consistent reliance on premium-cost labor. The system routinely uses travel nurses, agency staff, and overtime to meet baseline operational needs—not just to cover seasonal peaks or unexpected absences. This represents a sustained attempt to purchase more of the constrained resource at premium prices.[3]

[3] Hospital labor expenses increased 37% per patient between 2019 and 2022. Median hourly wages for contract nurses rose 106% during this period (from $64 to $132 per hour), while wages for employed nurses increased only 11%. Contract labor rose from 2% to 11% of total labor expenses. See Kaufman Hall, "National Hospital Flash Report," May 2022. https://www.kaufmanhall.com/insights/research-report/national-hospital-flash-report-may-2022

Service line growth limited by staffing. Strategic plans for service line expansion cite staffing as the primary barrier, not facility space, equipment, or physician availability. Approved capital projects sit idle awaiting staff to operate them.

Discharge delays. Patients clinically ready for discharge remain in beds because nursing tasks (education, medication reconciliation, coordination) are delayed due to competing demands on nursing time.

Elevated overtime rates. Overtime consistently runs above target levels, indicating that scheduled staffing is insufficient to meet demand.

None of these signs alone confirms a nursing constraint. But a system exhibiting most or all of them would be operating in a manner consistent with nursing as the binding limitation on throughput.

5 What You Would Expect to See: Financial Paradoxes

Beyond operational signs, a nursing-constrained system would exhibit certain financial patterns that might puzzle leadership if the constraint is not recognized.

Capacity investments that don't improve margins. The system invests in additional beds, new operating rooms, or expanded facilities. Volume does not increase proportionally because nursing staff are not available to support the new capacity. The investment generates depreciation expense without corresponding revenue growth.

High-margin service line growth that doesn't improve the bottom line. A service line with excellent contribution margin expands, but overall system margins remain flat or decline. The expansion consumed nursing capacity that was previously supporting other services. The "profitable" growth displaced even more profitable throughput elsewhere.

Persistent labor cost pressure despite volume stability. Even without volume growth, nursing labor costs rise year over year as the system pays premium rates to maintain current throughput. The constraint is increasingly expensive to feed.

Service mix changes that should improve margins but don't. Strategic initiatives shift case mix toward higher-margin services, but financial results disappoint. The higher-margin services consume more nursing hours per case, reducing total case volume the system can support.

These paradoxes have explanations within the constraint framework. Without that framework, they appear as execution failures, market headwinds, or simply bad luck.

6 Why Conventional Accounting Doesn't Reveal This

Hospital cost accounting systems are sophisticated tools built over decades to meet real needs: Medicare cost reporting, payer negotiations, departmental budgeting, and capital planning. They do these jobs well.

But they were not designed to answer a different question: *When a critical resource is constrained, which services make the best use of that resource?*

This is not a criticism of the systems or the people who built and use them. It is recognizing that different questions require different analytical lenses.

6.1 How Nursing Costs Are Treated

In most hospital accounting systems, nursing services are bundled into daily room charges alongside housekeeping, dietary, and other routine services. A patient requiring twelve hours of nursing care per day pays the same room rate as one requiring four hours. The variation in nursing intensity is invisible to the billing and cost accounting systems.

When nursing costs are allocated to service lines, they typically flow through as part of overhead—spread across services using allocation bases like patient days, charges, or relative value units. A service that consumes heavy nursing resources and one that consumes light nursing resources may receive similar nursing cost allocations if they have similar charges or lengths of stay.

The result: service line profitability reports do not reveal which services consume disproportionate nursing hours relative to their financial contribution.

6.2 The Averaging Problem

Cost-to-charge ratios and step-down allocation methods produce averaged costs. Industry research has documented cases where patients with identical diagnoses showed actual costs varying by more than 70%—but cost accounting showed them as identical.[4]

When costs are averaged, the signal about constraint consumption is lost. High-nursing-intensity cases are undercosted; low-intensity cases are overcosted. Strategic decisions based on these averaged costs may systematically favor the wrong services.

6.3 No Villain in This Story

It is worth emphasizing: these methods evolved for legitimate purposes and represent genuine expertise. Medicare cost reporting requirements shaped hospital accounting practices. The complexity of hospital operations resists simple costing approaches. The people who developed and maintain these systems are skilled professionals solving difficult problems.

The issue is not that the systems are wrong. It is that they were designed to answer certain questions, and the constraint question is not among them. Recognizing this gap is the first step toward addressing it.

6.4 Industry Recognition of Cost Accounting Limitations

The healthcare industry itself has begun to recognize that traditional cost accounting methods may be inadequate for strategic decision-making. In 2019, the Healthcare Financial Management Association (HFMA)—the nation's premier membership organization for healthcare finance leaders—partnered with Strata Decision Technology to develop the HFMA-Strata L7 Cost Accounting Maturity Model.[5]

The L7 Model provides a framework for assessing cost accounting sophistication across seven levels. At the base (Level 1), organizations use ratio of cost-to-charge methods—essentially spreading costs

[4] Kaplan, Robert S. et al. "Using Time-Driven Activity-Based Costing to Identify Value-Improvement Opportunities in Healthcare." *Journal of Healthcare Management* 59, no. 6 (November–December 2014): 399–413. https://www.hbs.edu/faculty/Pages/item.aspx?num=48462

[5] "Healthcare's First Cost Accounting Adoption Model Launched by HFMA and Strata Decision Technology," HFMA Press Release, February 11, 2019. https://www.hfma.org/press-releases/healthcares-first-cost-accounting-adoption-model-launched-by-hfma-and-strata-decision-technology/

using charge data as a proxy for resource consumption. At Level 7, organizations employ time-driven activity-based costing across all clinical areas, tracking actual resource consumption at the patient level.

The industry distribution is striking. According to surveys conducted at the model's launch, approximately 90 percent of hospitals and health systems lacked advanced cost accounting systems capable of providing accurate cost data across the continuum of care.[6] Less than 10 percent of healthcare delivery systems had achieved advanced cost accounting capabilities (Levels 5–7).

Harvard Business School professors Robert Kaplan and Michael Porter have been among the most prominent advocates for more accurate cost measurement in healthcare. Their work on time-driven activity-based costing (TDABC) demonstrated that traditional hospital costing methods can misstate actual costs by 50% or more at the patient level.[7]

The relevance to the nursing constraint hypothesis is direct: if 90 percent of hospitals cannot accurately measure costs at the service line or patient level, they certainly cannot measure which services consume disproportionate amounts of nursing hours relative to their contribution. The data required to identify a nursing constraint simply does not exist in most hospital cost accounting systems.

7 How to Test the Hypothesis

The preceding sections describe what a nursing-constrained system would look like. But how would a health system determine whether nursing is actually its binding constraint?

The author offers the following as candidate approaches, not proven methodologies. They represent the author's best current thinking on how one might test this hypothesis. Feedback and alternative approaches are welcome.

7.1 Capacity Utilization Analysis

Calculate utilization rates for each potentially constrained resource: nursing FTEs (hours worked versus hours available), inpatient beds, operating rooms, major equipment (cath labs, MRI, etc.), and key physician specialties.

The resource closest to or exceeding 100% utilization is likely the constraint. If nursing hours are consistently at or above capacity while beds show 75% occupancy and ORs show 80% utilization, nursing is the probable bottleneck.

A refinement: examine utilization by shift and unit. Constraints often manifest at specific times (day shift) or locations (ICU, med-surg) before appearing system-wide.

7.2 Queue Analysis

Identify where work accumulates and waits. In a nursing-constrained system, you would expect to see:

- Patients waiting in ED for inpatient beds (when beds are physically available but unstaffed)

[6] "Health Care's First Cost Accounting Adoption Model Is Now Available," HFMA, February 15, 2019. https://www.hfma.org/accounting-and-financial-reporting/cost-of-care/63271/

[7] Kaplan, Robert S. and Michael E. Porter. "How to Solve the Cost Crisis in Health Care." *Harvard Business Review*, September 2011. https://hbr.org/2011/09/how-to-solve-the-cost-crisis-in-health-care

- Patients waiting in PACU for floor beds (same reason)
- Discharge orders written but patients not discharged (nursing tasks incomplete)
- Procedures delayed for nursing coverage rather than surgeon or equipment availability

If queues consistently form at nursing-dependent handoffs rather than at physical capacity limits or physician availability, nursing is likely the constraint.

7.3 The Marginal Resource Test

Ask the thought experiment: "If we could add one more nurse tomorrow, would we treat more patients?"

If the answer is yes—if adding nursing capacity would immediately translate to additional throughput—nursing is likely the constraint. If the answer is no (because beds are full, ORs are booked, or physicians are unavailable), the constraint lies elsewhere.

This can be tested empirically by examining what happens during periods of better-than-usual staffing. Does throughput increase? Do bottlenecks resolve? If so, nursing capacity was the limitation.

7.4 Throughput-per-Nursing-Hour Calculation

For each major service line, calculate:

- Average contribution margin per case (revenue minus truly variable costs)
- Average nursing hours per case (direct care hours, not allocated overhead)
- Contribution margin per nursing hour

Compare the ranking by margin-per-case with the ranking by margin-per-nursing-hour. If the rankings differ substantially, the constraint lens would lead to different strategic conclusions than conventional analysis.

This calculation requires data that may not be readily available from standard cost accounting systems. Nursing hours per case may need to be estimated from staffing patterns and length of stay, or derived from time studies or acuity systems if available.

7.5 Behavioral Indicators

Sometimes the system's behavior reveals the constraint more clearly than formal analysis. Two indicators worth noting:

Active nursing recruitment. If the organization is actively recruiting to fill many nursing positions—not just normal turnover replacement but sustained, high-volume hiring efforts—it has implicitly identified nursing as a capacity limitation. The system is attempting to elevate the constraint even without formal constraint analysis. Job postings, signing bonuses, and recruitment campaigns are observable signals.

Pressure on existing nursing capacity. If nurses are reporting pressure to forgo accrued time off, work overtime, or delay vacations, the system is attempting to extract more capacity from its constrained resource. Union grievances, staff surveys, or informal complaints about such pressure are evidence that demand for nursing hours exceeds supply. The system is behaviorally revealing what it may not have formally articulated.

These indicators have the virtue of being observable without access to sophisticated financial or operational data. A reader could assess them from general organizational awareness, public job postings, or union communications.

7.6 Limitations and Uncertainties

The author acknowledges significant uncertainty in these approaches:

- Nursing hours per case are not routinely tracked in most cost accounting systems; estimates may be rough
- Multiple constraints may bind simultaneously at different points in the system
- The constraint may shift based on patient mix, time of day, or seasonal factors
- Defining "truly variable costs" for contribution margin calculation involves judgment calls

These approaches are offered as starting points for investigation, not as validated diagnostic tools. The author would welcome input from those with deeper operational and financial expertise on how to refine or replace them.

8 Implications If True

If nursing hours are indeed the binding constraint, several implications follow for strategic decision-making.

8.1 Service Line Evaluation Would Change

Instead of asking "Which service lines have the best margins?" leadership would ask "Which service lines generate the most contribution per nursing hour consumed?" Services that look marginal under conventional analysis might be highly valuable if they use nursing hours efficiently. Services that look excellent might be revealed as poor uses of constrained capacity.

This does not mean abandoning high-margin services. It means understanding their true cost in terms of constraint consumption and making explicit tradeoffs rather than invisible ones.

8.2 Investment Priorities Would Shift

Investments that relieve the nursing constraint would take priority over investments that add other capacity. This might include:

- Staffing investments (competitive compensation, retention programs, residency programs)
- Technology that increases nursing productivity (workload reduction, documentation efficiency)
- Process changes that reduce nursing time per patient (care model redesign, discharge optimization)
- Support staff that can absorb non-nursing tasks currently performed by nurses

Investments that add beds, ORs, or equipment without addressing the nursing constraint would be deprioritized—they add capacity the system cannot use.

8.3 "Exploiting the Constraint" Would Become a Priority

In constraint terminology, "exploiting" the constraint means ensuring not a single unit of constraint capacity is wasted. For nursing, this would mean:

- Minimizing non-nursing tasks performed by nurses
- Reducing documentation burden and administrative overhead
- Ensuring smooth patient flow to avoid nurses waiting for beds, equipment, or physicians
- Aggressive management of discharge timing to free nursing capacity for new admissions

Every hour of nursing time spent on tasks that could be done by others, or spent waiting due to system inefficiencies, is lost throughput.

8.4 The Math Becomes Explicit

Perhaps most importantly, the tradeoffs become visible. When leadership considers expanding a nursing-intensive service, they would explicitly calculate: How many nursing hours will this consume? What throughput are we giving up elsewhere? Is the margin on the new service sufficient to justify the constraint consumption?

These questions are not currently part of most strategic planning processes because the data to answer them is not routinely produced. Making the math explicit would change how decisions are made.

8.5 What This Framework Does Not Mean

A note of caution is warranted. Constraint-based thinking is a lens for understanding tradeoffs—not a replacement for clinical judgment, ethical obligations, or community mission.

Emergency care is not discretionary. Obstetric coverage may be a community obligation regardless of nursing intensity. Trauma services exist because someone must provide them. Equity and access commitments constrain which services a health system can decline to offer.

The framework described here would not tell a hospital to stop providing emergency services because they consume nursing hours inefficiently. It might help leadership understand why expanding elective high-acuity programs while emergency volume grows creates unsustainable pressure—and why investments in nursing capacity or emergency department efficiency might have higher strategic value than they appear under conventional analysis.

Throughput accounting is a decision aid within constraints, not the goal itself. The goal remains serving patients and communities. The framework simply asks whether current analytical tools are revealing or obscuring the tradeoffs inherent in that service.

9 An Invitation

This paper presents a hypothesis: that nursing hours may be an unrecognized or underappreciated constraint in many health systems, and that conventional cost accounting obscures this reality. The hypothesis has not been tested against real health system data. The author does not know whether it is correct.

What the author does believe is that the question is worth asking. The nursing workforce challenge is real and persistent. The financial pressures on health systems are intense. If the analytical

frameworks currently in use are systematically pointing toward the wrong strategic decisions, that would be important to know.

The author invites feedback from health system leaders, CFOs, CNOs, and operational experts:

- Does the operational picture described here match your experience?
- Are there flaws in the reasoning that would invalidate the hypothesis?
- What data would you need to see to evaluate whether nursing is your system's constraint?
- Are there better approaches to testing this hypothesis than those suggested here?

If the hypothesis is wrong, understanding why it is wrong would be valuable. If it has merit, exploring how to apply constraint-based thinking to health system strategy could be valuable as well.

This is not a sales pitch. It is an invitation to think together about a question that may matter.

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Related Work

This paper is part of a series exploring applications of systems thinking to healthcare operations:

Improving Health Systems At Scale: Automated Policy Conflict Analysis. Introduces a capability for systematic identification of embedded conflicts in healthcare policy documents using AI-powered analysis grounded in Theory of Constraints methodology. Available from the author.

Theory of Constraints Analysis for Healthcare Policy Conflict Identification. Presents the methodology for systematic policy conflict analysis in detail, with a worked example demonstrating application to end-of-life care policies. Available from the author.

The Hazard We Don't See: How Policy Conflicts Harm the People Organizations Are Meant to Protect. Examines the relationship between embedded policy conflicts and organizational safety failures. Available from the author.

Post-Acute Care Plan (PACP) Concept. Proposes a proactive approach to discharge planning that could significantly reduce average length of stay for complex discharge cases. Available from the author.

About the Author

John Sambrook is President of Common Sense Systems, Inc., a consulting firm he founded in 1996. His background includes thirty years of embedded systems engineering and medical device development, including work at SonoSite on portable ultrasound systems.

His interest in constraint-based thinking traces to study with Dr. Eliyahu Goldratt, who originated the Theory of Constraints body of knowledge. The framework's applicability to healthcare has become increasingly apparent as health systems face simultaneous pressure on margins and workforce availability.

This paper represents an attempt to apply constraint-based thinking to healthcare strategy and an invitation for those with deeper domain expertise to evaluate whether the framework offers useful insight.

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